



McLaren
USER MANUAL
MCLAREN 570S GT4



TABLE OF CONTENTS

CLICK TO VIEW A SECTION

GENERAL INFORMATION

<i>A Message From iRacing »</i>	3
<i>Tech Specs »</i>	4
<i>Introduction »</i>	5
<i>Getting Started »</i>	5
<i>Loading An iRacing Setup »</i>	6
<i>Dash Pages »</i>	7

ADVANCED SETUP OPTIONS

<i>Tires & Aero »</i>	9
<i>Tire Data »</i>	10
<i>Chassis »</i>	11
<i>Front »</i>	11
<i>In-Car Dials »</i>	12
<i>Front Corners »</i>	13
<i>Rear Corners »</i>	15
<i>Rear »</i>	16

SETUP TIPS

<i>Setup Tips »</i>	18
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McLaren

DEAR iRACING USER,

Dear iRacing User,

Congratulations on your purchase of the McLaren 570s GT4! From all of us at iRacing, we appreciate your support and your commitment to our product. We aim to deliver the ultimate sim racing experience, and we hope that you'll find plenty of excitement with us behind the wheel of your new car!

Few manufacturers can match the thrill of driving a McLaren in any class or specification, and the 570S, the legendary brand's foray into the world of GT4s, is no exception. Powered by a 3.8-liter twin turbo V8 engine and sitting on McLaren's carbon fiber MonoCell II chassis, this highly agile machine is the car of choice for a number of customer race teams competing in GT4 formula series around the world.

The 570S GT4 broke cover at the 2016 Goodwood Festival of Speed and made its competitive debut in British GT under the brand's own supervision that same year, allowing customers that took delivery of the cars in 2017 to drive a racer that had been refined in real competition. Since then, it's proven to be a highly popular car with anyone who's driven it, setting a high water mark of 140 global podium finishes in 2019.

The following guide explains how to get the most out of your new car, from how to adjust its settings off of the track to what you'll see inside of the cockpit while driving. We hope that you'll find it useful in getting up to speed.

Thanks again for your purchase, and we'll see you on the track!



CHASSIS

**MCLAREN CARBON FIBER MONOCELL
II CHASSIS WITH INDEPENDENT FRONT
AND REAR COILOVER SUSPENSION**



LENGTH
4606 mm
181.3 in

WIDTH
2095 mm
82.5 in

WHEELBASE
2674 mm
105.3 in

DRY WEIGHT
1425 kg
3142 lbs

WET WEIGHT
WITH DRIVER
1540 kg
3395 lbs

POWER UNIT



TWIN TURBO V8 M838TE

DISPLACEMENT
3.8 Liters
231.9 CID

RPM LIMIT
7450 RPM

TORQUE
355 lb-ft
481 Nm

POWER
450 bhp
336 kW



INTRODUCTION

The information found in this guide is intended to provide a deeper understanding of the chassis setup adjustments available in the garage, so that you may use the garage to tune the chassis setup to your preference.

Before starting the car, it is recommended to map controls for Brake Bias and ABS/TCS/ESC settings. While this is not mandatory, this will allow you to make quick changes to the brake bias and stability management systems to suit your driving while out on track.

Once you load into the car, getting started is as easy as pulling the “upshift” paddle to put it into gear, and hitting the accelerator pedal. This car uses an automated sequential transmission and does not require manual clutch operation to shift in either direction.

However, the car’s downshift protection will not allow you to downshift if it feels you are traveling too fast for the gear requested. If that is the case, the downshift command will simply be ignored.

Upshifting is recommended when the shift lights on the dashboard are all fully illuminated. This is between 6800 and 7300 rpm depending on the currently selected gear.

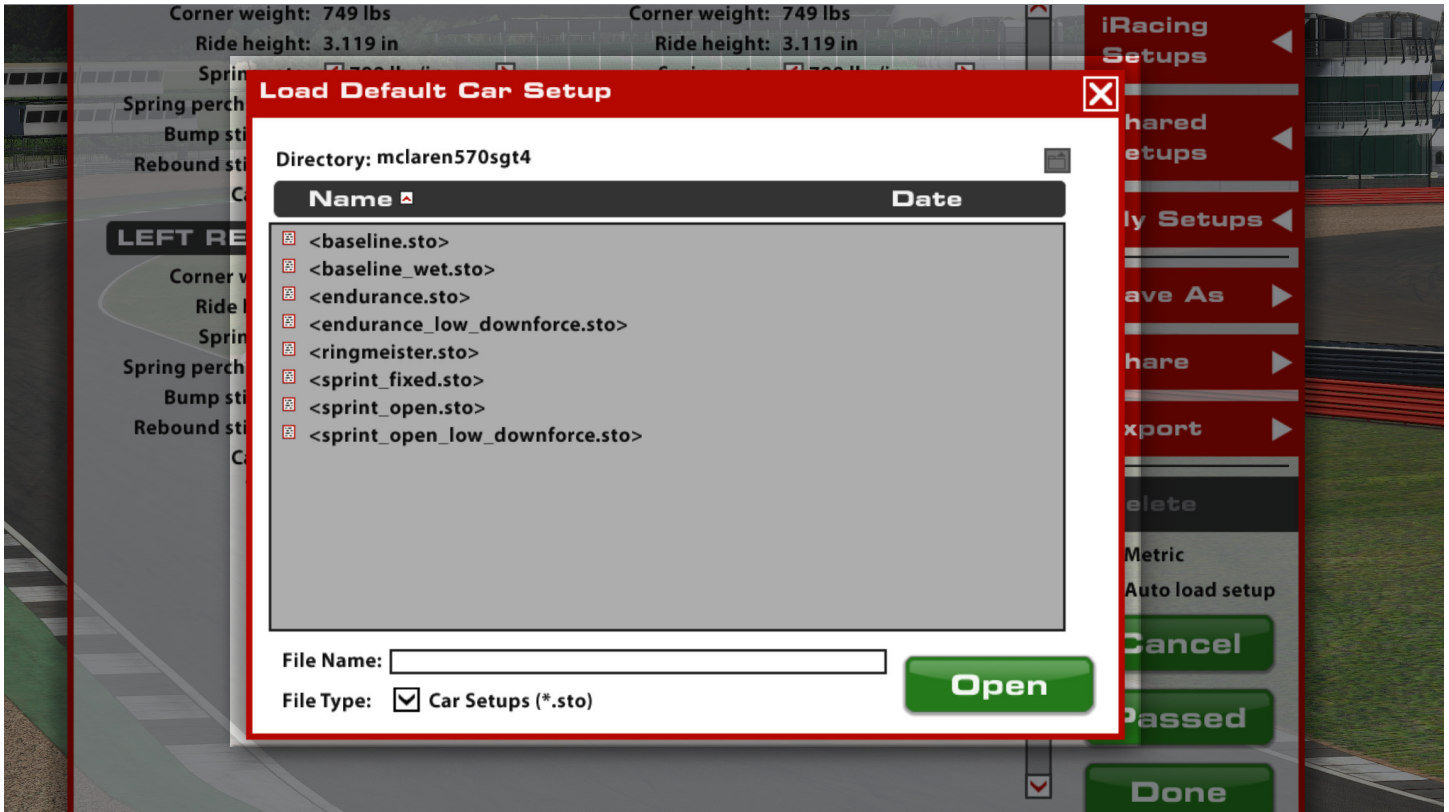
GETTING STARTED



Once you load into the car, getting started is as easy as selecting the “upshift” button to put it into gear, and hitting the accelerator pedal. This car uses a sequential transmission and does not require a clutch input to shift in either direction.

However the car’s downshift protection will not allow you to downshift if it feels you are traveling too fast for the gear selected and would incur engine damage. If that is the case, the gear change command will simply be ignored.

LOADING AN iRACING SETUP



Upon loading into a session, the car will automatically load the iRacing Baseline setup [baseline.sto]. If you would prefer one of iRacing's pre-built setups that suit various conditions, you may load it by clicking Garage > iRacing Setups > and then selecting the setup to suit your needs.

If you would like to customize the setup, simply make the changes in the garage that you would like to update and click apply.

If you would like to save your setup for future use click "Save As" on the right to name and save the changes. To access all of your personally saved setups, click "My Setups" on the right side of the garage.

If you would like to share a setup with another driver or everyone in a session, you can select "Share" on the right side of the garage to do so.

If a driver is trying to share a setup with you, you will find it under "Shared Setups" on the right side of the garage as well.

DASH CONFIGURATION

The dash display in this car is non-adjustable and features a single page to display critical vehicle information.

RACE DASH CONFIGURATION



Front Dash	
Rear Dash row 1 left	Graphical gain/loss lap time depiction relative to session best lap
Front Dash row 1 right	Battery voltage indicator
Front Dash center	Graphical depiction of engine rpm
Front Dash row 2 left	Engine water temperature (Celsius or Fahrenheit)
Front Dash row 2 center	Currently selected gear
Front Dash row 2 right upper	Last lap time as m:ss:ms
Front Dash row 2 right lower	gain/loss lap time as ss:ms relative to session best lap
Front Dash row 3 left	Engine oil pressure (Bar or psi)
Front Dash row 3 right	Fuel pressure (Bar or psi)

Rear Dash	
Rear Dash row 1 left	Total vehicle distance traveled (km or miles)
Rear Dash row 2 left	Distance traveled in current session (km or miles)
Rear Dash row 3 left	Distance traveled since last pitstop (km or miles)
Rear Dash vehicle graphic tires	Tire pressure indicators for each tire
Rear Dash vehicle graphic top	Engine oil level indicator
Rear Dash vehicle graphic center	Engine water temperature indicator
Rear Dash vehicle graphic bottom	Engine oil temperature indicator
Rear Dash lower right fill bar	Graphical depiction of current fuel level along with percentage.
Rear Dash lower right	Outside air temperature (Celsius or Fahrenheit)
Rear Dash far right	ESC Status, off = Normal, Orange = Sport, Orange with 'off' = Track
Rear Dash far right bottom	ABS activation indicator, illuminates orange when ABS is active

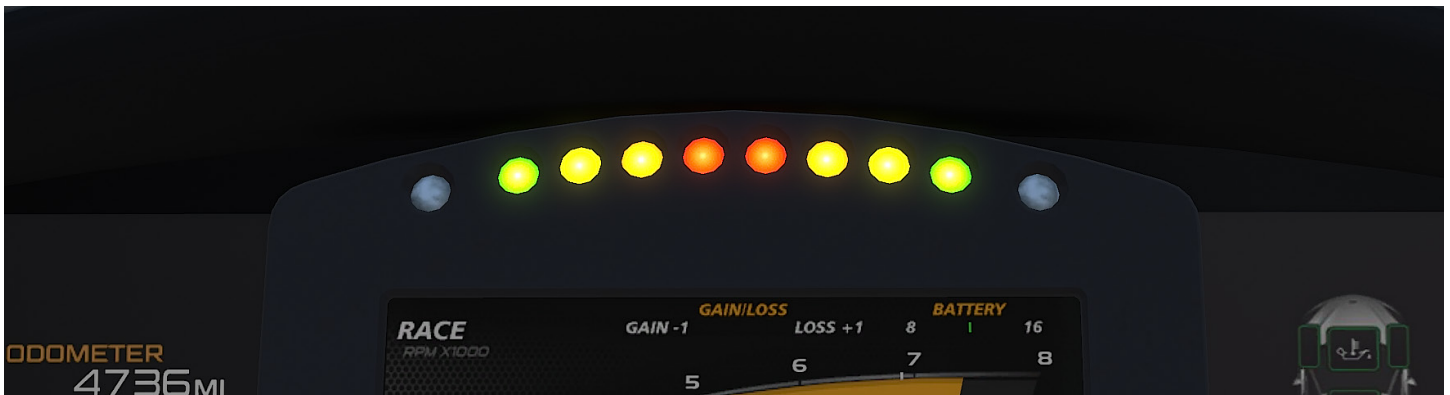


PIT SPEED LIMITER



When the pit limiter is active a large blue box containing 'PIT LIMITER ON' will appear at the base of the forward dashboard along with the illumination of the two centermost shift light LED's.

SHIFT LIGHTS



The shift lights illuminate from the outer edges inwardly. Illumination of the first LED will shift depending on the selected gear as such, the below values are only valid for 3rd gear.

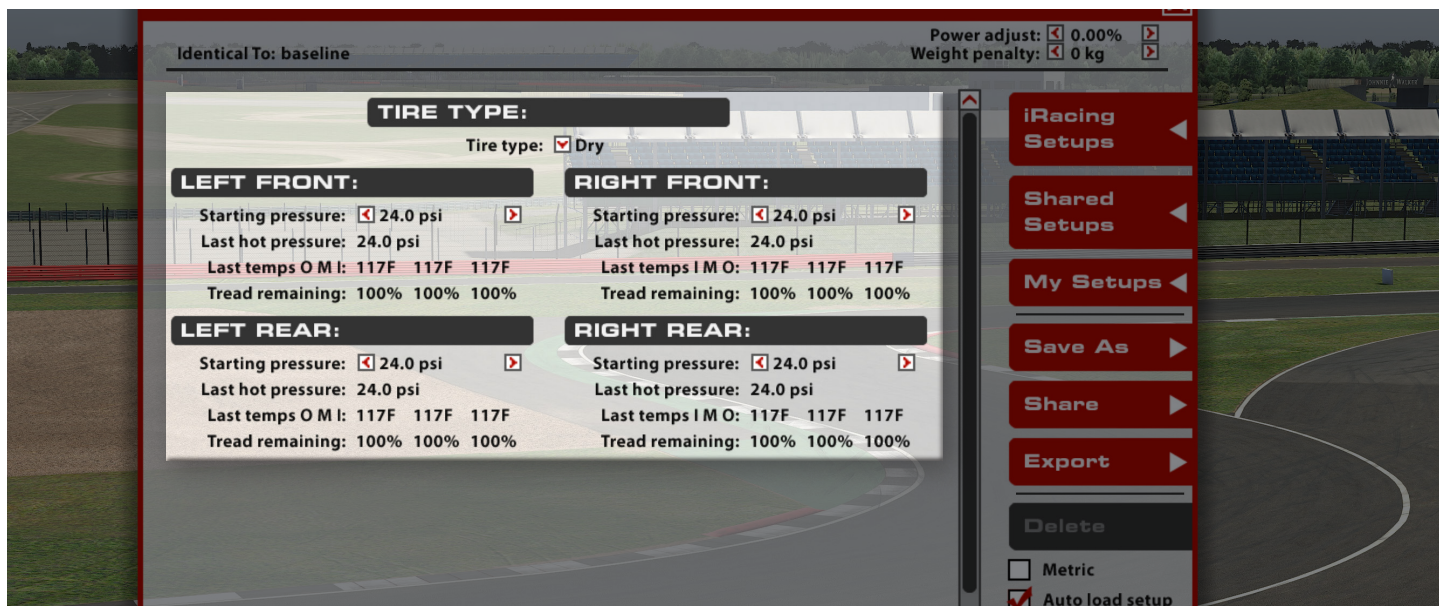
- 2 Green - 6000 rpm
- 2 Yellow - 6300 rpm
- 4 Yellow - 6600 rpm
- 2 Red - 6900 rpm

ADVANCED SETUP OPTIONS

This section is aimed toward more advanced users who want to dive deeper into the different aspects of the vehicle's setup. Making adjustments to the following parameters is not required and can lead to significant changes in the way a vehicle handles. It is recommended that any adjustments are made in an incremental fashion and only singular variables are adjusted before testing changes.

TIRES & AERO

TIRE DATA



TIRE TYPE

Selects which type of tire is installed on the car when loaded into the world. Dry, or slick, tires are used for dry racing conditions while Wet tires are intended for raining and wet track conditions.

COLD AIR PRESSURE

Air pressure in the tire when the car is loaded into the world. Higher pressures will reduce rolling drag and heat buildup, but will decrease grip. Lower pressures will increase rolling drag and heat buildup, but will increase grip. Higher speeds and loads require higher pressures, while lower speeds and loads will see better performance from lower pressures. Cold pressures should be set to track characteristics for optimum performance. Generally speaking, it is advisable to start at lower pressures and work your way upwards as required.

HOT AIR PRESSURE

Air pressure in the tire after the car has returned to the pits. The difference between cold and hot pressures can be used to identify how the car is progressing through a run in terms of balance, with heavier-loaded tires seeing a larger difference between cold and hot pressures. Ideally, tires that are worked in a similar way should build pressure at the same rate to prevent a change in handling balance over the life of the tire, so cold pressures should be adjusted to ensure that similar tires are at similar pressures once up to operating temperature. Hot pressures should be analyzed once the tires have stabilized after a period of laps. As the number of laps per run will vary depending upon track length a good starting point is approximately 50% of a full fuel run.

TIRE TEMPERATURES

Tire carcass temperatures, measured via Pyrometer, once the car has returned to the pits. Wheel Loads and the amount of work a tire is doing on-track are reflected in the tire's temperature, and these values can be used to analyze the car's handling balance. Center temperatures are useful for directly comparing the work done by each tire, while the Inner and Outer temperatures are useful for analyzing the wheel alignment (predominantly camber) while on track. These values are measured in three zones across the tread of the tire. Inside, Middle and Outer.

TREAD REMAINING

The amount of tread remaining on the tire once the car has returned to the pits. Tire wear is very helpful in identifying any possible issues with alignment, such as one side of the tire wearing excessively, and can be used in conjunction with tire temperatures to analyze the car's handling balance. These values are measured in the same zones as those of temperature.

CHASSIS

FRONT

Identical To: baseline Weight penalty: 0 kg

FRONT:

ARB setting: 1

Toe-in: -8/64"

Cross weight: 50.0%

LEFT FRONT:

Corner weight: 749 lbs

Ride height: 3.119 in

Spring rate: 799 lbs/in

Spring perch offset: 6.693"

Bump stiffness: +13 clicks

Rebound stiffness: +7 clicks

Camber: -4.4 deg

LEFT REAR:

Corner weight: 949 lbs

Ride height: 4.044 in

Spring rate: 1299 lbs/in

Spring perch offset: 5.551"

IN-CAR DIALS:

Brake pressure bias: 54.0%

Brake pads: Medium friction

ABS / TC / ESC Setting: Track

RIGHT FRONT:

Corner weight: 749 lbs

Ride height: 3.119 in

Spring rate: 799 lbs/in

Spring perch offset: 6.693"

Bump stiffness: +13 clicks

Rebound stiffness: +7 clicks

Camber: -4.4 deg

RIGHT REAR:

Corner weight: 949 lbs

Ride height: 4.044 in

Spring rate: 1299 lbs/in

Spring perch offset: 5.551"

ARB SETTING

Increasing the ARB setting shortens the ARB moment arm and will increase the roll stiffness of the front suspension, resulting in less body roll but increasing mechanical understeer. This can in some cases, lead to a more responsive steering feel for the driver. Conversely, reducing the ARB setting lengthens the ARB moment arm, softening the suspension in roll and increasing body roll but decreasing mechanical understeer. This can result in a less-responsive feel from the steering, but grip across the front axle will increase. Along with this, the effects of softening or stiffening the ARB assembly in relation to aerodynamics should also be considered, a softer ARB configuration will result in more body roll which will decrease control of the aero platform in high speed corners and potentially lead to a loss in aero efficiency. Two ARB settings are available; 1 is 'soft' and 2 is 'stiff'.

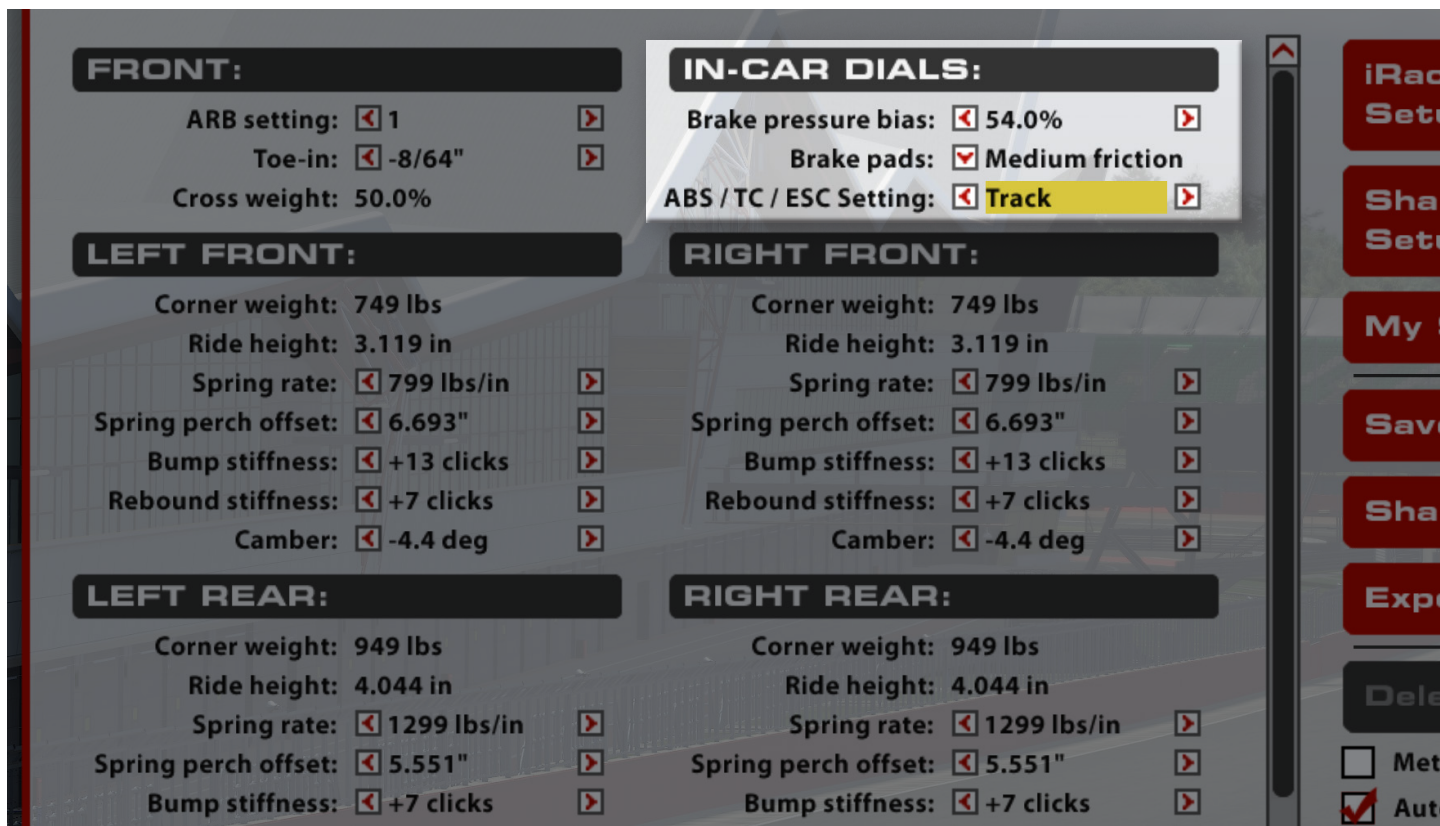
TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and Toe-out is the opposite. On the front end, adding toe-out will increase slip in the inside tire and decrease straight-line stability while adding toe-in will reduce the slip and increase straight-line stability.

CROSS WEIGHT

The percentage of total vehicle weight in the garage acting across the right front and left rear corners. 50.0% is generally optimal for non-oval tracks as this will produce symmetrical handling in both left and right hand corners providing all other chassis settings are symmetrical. Higher than 50% cross weight will result in more understeer in left hand corners and increased oversteer in right hand corners, cross weight can be adjusted by making changes to the spring perch offsets at each corner of the car.

IN-CAR DIALS



BRAKE PRESSURE BIAS

Brake Bias is the percentage of braking force that is being sent to the front brakes. Values above 50% result in greater pressure in the front brake line relative to the rear brake line which will shift the brake balance forwards increasing the tendency to lock up the front tyres but potentially increasing overall stability in braking zones. This should be tuned for both driver preference and track conditions to get the optimum braking performance for a given situation.

BRAKE PADS

The vehicle's braking performance can be altered via the Brake Pad Compound. The "Low" setting provides the least friction, reducing the effectiveness of the brakes but providing the most modulation, while "Medium" and "High" provide more friction and increase the effectiveness of the brakes but the least modulation.

ABS/TC/ESC SETTING

This option provides a combined control for the traction control (TC) and ABS (Anti-Lock Braking System). Three options are available: 'Normal' which provides maximum TC and ABS assistance, 'Sport' which provides a reduced level of TC and ABS assistance and 'Track' which provides the lowest levels of TC and ABS intervention. 'Track' is the recommended setting.

FRONT CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the spring perch offset adjustments at each corner.

FRONT RIDE HEIGHT

Distance from ground to a reference point on the chassis. Since these values are measured to a specific reference point on the car, these values may not necessarily reflect the vehicle's ground clearance, but instead provide a reliable value for the height of the car off of the race track at static values. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip. Increasing front ride height will decrease front downforce as well as decrease overall downforce, but will allow for more weight transfer across the front axle when cornering. Conversely, reducing ride height will increase front and overall downforce, but reduce the weight transfer across the front axle.

SPRING RATE

This setting determines the installed corner spring stiffness. Stiffer springs will result in a smaller variance in ride height between high and low load cases and will produce superior aerodynamic performance through improved platform control; however, they will also result in increased tire load variation which will manifest as a loss in mechanical grip. Typically the drawbacks of stiffer springs will become more pronounced on rougher tracks and softer springs in these situations will result in increased overall performance. Corner spring changes will influence both roll and pitch control of the platform and ARB changes should be considered when altering corner spring stiffnesses in order to retain the same front to rear roll stiffness and overall balance. When reducing corner spring stiffness the ARB stiffness should be increased to retain the same roll stiffness as previously. Spring perch offsets must be adjusted to return the car to the prior static ride heights after any spring rate change.

SPRING PERCH OFFSET

Used to adjust the ride height at this corner of the car by changing the installed position of the spring. Increasing the spring perch offset will result in lowering this corner of the car while reducing the spring perch offset will raise this corner of the car. These changes should be kept symmetrical across the axle (left to right) to ensure the same corner ride heights and no change in cross weight. The spring perch offsets can also be used in diagonal pairs (LF to RR and RF to LR) to change the static cross weight in the car.

BUMP STIFFNESS

The bump stiffness setting is a paired adjustment controlling both the low and high speed compression damping characteristics of the damper. In this case 0 is minimum damping (least resistance to compression) while 18 is maximum damping (most resistance to compression). Increasing the bump stiffness will result in a faster transfer of weight to this corner of the car during transient movements such as braking and direction change with increased damping usually providing an increase in turn-in response but a reduction in overall grip in the context of front dampers. High speed compression damping will increase proportionally to the increase in low speed compression damping which will also result in harsher response to kerb strikes. At smoother tracks more bump stiffness will typically increase performance while at rougher tracks or ones with aggressive kerbs less compression damping can result in an increase in mechanical grip at the expense of platform control.

FRONT CORNERS

REBOUND STIFFNESS

The Rebound Stiffness setting is a paired adjustment to both low and high speed rebound damping characteristics. Increasing rebound damping will slow down the rate at which the damper extends in both low and high speed situations. A typical low damper speed situation would be as the car rolls back to level on a corner exit while a high speed situation would be where the suspension is extending after large kerb contact. 0 is minimum damping (least resistance to extension) while 18 is maximum damping (most resistance to extension). While high rebound stiffness will result in improved platform control for aerodynamic performance and overall chassis response it is important to avoid situations where the shock is too slow in rebounding as this will result in the tire losing complete contact with the track surface which can induce or exacerbate severe oscillations.

CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Due to suspension geometry and corner loads, negative camber is desired on all four wheels. Higher negative camber values will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance. Increasing front camber values will typically result in increased front axle grip during mid to high speed cornering but will result in a loss of braking performance and necessitate a rearward shift in brake bias to compensate.

REAR CORNERS

LEFT REAR:		RIGHT REAR:	
Corner weight:	749 lbs	Corner weight:	749 lbs
Ride height:	3.119 in	Ride height:	3.119 in
Spring rate:	799 lbs/in	Spring rate:	799 lbs/in
Spring perch offset:	6.693"	Spring perch offset:	6.693"
Bump stiffness:	+13 clicks	Bump stiffness:	+13 clicks
Rebound stiffness:	+7 clicks	Rebound stiffness:	+7 clicks
Camber:	-4.4 deg	Camber:	-4.4 deg
Toe-in:	+1/1000"	Toe-in:	+1/1000"

REAR: Fuel level: 14.5 gal

REAR RIDE HEIGHT

Distance from ground to a reference point on the rear of the chassis. Increasing rear ride height will decrease rear downforce as well as increase overall downforce and will allow for more weight transfer across the rear axle when cornering. Conversely, reducing ride height will increase rear downforce percentage but reduce overall downforce while reducing the weight transfer across the rear axle. Rear ride height is a critical tuning component for both mechanical and aerodynamic balance considerations and static rear ride heights should be considered and matched to the chosen rear corner springs for optimal performance.

SPRING RATE

Similar to at the front axle, stiffer springs will result in a smaller variance in ride height between high and low load cases and will produce superior aerodynamic performance through improved platform control at the expense of mechanical grip. This can be particularly prominent when exiting slow speed corners with aggressive throttle application. Stiffer springs will tend to react poorly during these instances especially so on rough tracks which will result in significant traction loss. Spring stiffness should be matched to the needs of the racetrack and set such that the handling balance is consistent between high and low speed cornering. As an example case, a car which suffers from high speed understeer but low speed oversteer could benefit from an increase in rear spring stiffness. This will allow for a lower static rear height which will reduce rear weight transfer during slow speed cornering while maintaining or even increasing the rear ride height in high speed cornering to shift the aerodynamic balance forwards and reduce understeer.

BUMP STIFFNESS

The bump stiffness setting is a paired adjustment controlling both the low and high speed compression damping characteristics of the damper with identical ranges to those of the front dampers. Increasing the compression damping will result in a faster transfer of weight to this corner of the car during transient movements such as accelerating and direction change with increased damping usually providing an increase in response but a reduction in overall grip especially at corner exit traction in the context of rear dampers. Excessively stiff compression damping can cause very poor traction on rough tracks as it can result in large tire load variation and a reduction in overall grip.

REBOUND STIFFNESS

The rebound stiffness setting is a paired adjustment controlling both the low and high speed damping characteristics of the damper with identical ranges to those of the front dampers. Increasing rebound damping will slow down the rate at which the damper extends in both low and high speed situations. As at the front, high rebound stiffness will result in improved platform control for aerodynamic performance and overall chassis response but it is important to avoid situations where the shock is too slow in rebounding as this will result in the tire losing complete contact with the track surface. This can be particularly detrimental during braking events and during the initial turn-in phase though an increase in rebound stiffness can help to 'slow down' the change in pitch of the car as the brakes are applied, potentially increasing braking stability.

REAR CORNERS

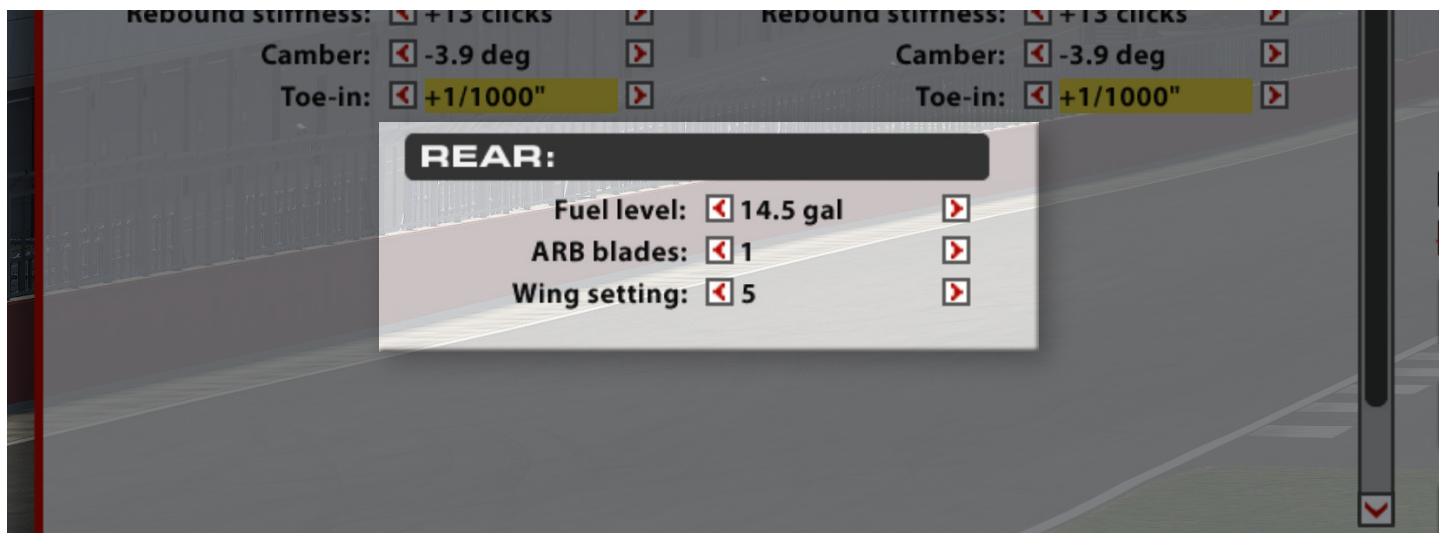
CAMBER

As at the front of the car it is desirable to run significant amounts of negative camber in order to increase the lateral grip capability; however, it is typical to run slightly reduced rear camber relative to the front. This is primarily for two reasons, firstly, the rear tires are wider compared to the fronts and secondly the rear tires must also perform the duty of driving the car forwards where benefits of camber to lateral grip become a tradeoff against reduced longitudinal (traction) performance.

TOE-IN

At the rear of the car it is typical to run toe-in. Increases in toe-in will result in improved straight line stability and a reduction in response during direction changes. Large values of toe-in should be avoided if possible as this will increase rolling drag and reduce straight line speeds. When making rear toe changes remember that the values are for each individual wheel as opposed to paired as at the front. This means that individual values on the rear wheels are twice as powerful as the combined adjustment at the front of the car when the rear toes are summed together. Generally, it is advised to keep the left and right toe values equal to prevent crabbing or asymmetric handling behaviour; however, heavily asymmetric tracks such as Lime Rock Park may see a benefit in performance from running asymmetric configurations of rear toe and other setup parameters.

REAR



FUEL LEVEL

The amount of fuel in the fuel tank when the car is loaded into the world.

ARB SETTING

Increasing the ARB assembly stiffness will increase the roll stiffness of the rear suspension, resulting in less body roll but increasing mechanical oversteer. This can also cause the car to “take a set” more quickly at initial turn-in. Conversely, reducing the ARB assembly stiffness will soften the suspension in roll, increasing body roll but decreasing mechanical oversteer. This can result in a less-responsive feel from the rear especially in transient movements, but grip across the rear axle will increase. Two ARB settings are available; 1 is ‘soft’ and 2 is ‘stiff’.

WING SETTING

The wing setting refers to the relative angle of attack of the rear wing, this is an aerodynamic device which has a significant impact upon the total downforce (and drag!) produced by the car as well as shifting the aerodynamic balance of the car rearwards with increasing angle. Increasing the rear wing angle results in more total cornering grip capability in medium to high speed corners but will also result in a reduction of straight line speed. Rear wing angle should be adjusted in conjunction with front and rear ride heights, specifically the difference between front and rear ride heights known as ‘rake’. To retain the same overall aerodynamic balance it is necessary to increase the rake of the car when increasing the rear wing angle.

SETUP TIPS

This section is aimed toward helping users who want to dive deeper into the different aspects of the vehicle's setup.



SETUP TIPS

In the iRacing Setup folder you will find several default setups to choose from:

BASELINE

50% fuel level, the default setup when first loading the car, intentionally conservative to allow you to acclimatize to a new car or track.

BASELINE_WET

50% fuel level, the default setup for a wet track with wet tires fitted and driver aids adjusted for wet conditions.

ENDURANCE

100% fuel level, intended for most tracks where optimal wing angles will be in the 3-5 range and race lengths are at, or over, one hour in length.

ENDURANCE LOW DOWNFORCE

100% fuel level, intended for tracks where straight-line performance is a priority (Daytona, Le Mans, etc.) and a low wing angle is required. Fueled for race lengths at, or over, one hour in length.

CHASSIS ADJUSTMENTS

Should you choose to make your own adjustments the easiest way to change the balance of the car is through the Anti-Roll Bars or the Wing setting:

Stiffer front ARB - More understeer

Stiffer rear ARB - More oversteer

Lower wing setting - More oversteer, less downforce, and higher straight-line speed.

Higher wing setting - More understeer, more downforce, and lower straight-line speed.

SPRINT_OPEN

56% fuel level, intended for most tracks in the Open Setup series where optimal wing angles are in the 3-5 range.

SPRINT_OPEN_LOW_DOWNFORCE

56% fuel level, intended for tracks in the Open Setup series where straight-line performance is a priority and optimal wing angles are low.

SPRINT_FIXED

50% fuel level, intended for short sprint races and the setup used for the Fixed Setup series.

In the event a setup fails tech inspection after fuel level or other changes it is likely the ride heights require adjustment. This is performed by using the Spring Perch Offset adjustments at either end of the car: Right-clicks (positive) will reduce the ride height, left-clicks (negative) will increase the ride height.